

A convex free-Gibbs counterexample to the weak free Cramer conjecture

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2026-07-18

Source and question

The source paper is Charles-Philippe Diez, *Free Stein kernels and moment maps*, arXiv:2410.02470. After recalling the Bercovici–Voiculescu failure of the ordinary free Cramer theorem, the source asks whether a weak version might hold inside the class of one-dimensional free Gibbs measures with strictly C^2 convex potentials:

Theorem 8. *Failure of Cramér’s theorem (Voiculescu, Bercovici, [7](#)).*

There exists $\mu, \nu \in \mathcal{P}(\mathbb{R})$ compactly supported probability measure on the real line which are not semicircular and such that their free convolution $\mu \boxplus \nu$ is semicircular.

However, we think that the result might be true thanks to a possible weak version of the free Cramer’s theorem, but unfortunately we have not been able to prove it, and not even the first proposition in this direction, which would be a free version of Prekopa’s theorem, asserting in the classical case that log-concavity is preserved (or strongly preserved) by the usual convolution.

Conjecture 1. *(Weak free Cramer’s conjecture) Let $\mu := \nu_\phi$ and $\nu := \nu_\psi$ be two free Gibbs measures associated with two strictly C^2 convex potentials $\phi, \psi : \mathbb{R} \rightarrow \mathbb{R}$. If the free convolution $\mu \boxplus \nu$ is semicircular, then both μ and ν are semicircular.*

In the notation of the source, the conjecture says that if $\mu = \nu_\phi$ and $\nu = \nu_\psi$ for strictly C^2 convex potentials $\phi, \psi : \mathbb{R} \rightarrow \mathbb{R}$, and if $\mu \boxplus \nu$ is semicircular, then both μ and ν are semicircular.

Proof intuition

Free additive convolution is linearized by the R -transform. The centered semicircle of variance 1 has $R(z) = z$. Thus, if one can realize two nonsemicircular free Gibbs laws with

$$R_+(z) = \frac{1}{2}z + az^2, \quad R_-(z) = \frac{1}{2}z - az^2,$$

then their free convolution is exactly semicircular, while the nonzero third free cumulants $\pm a$ prevent either factor from being semicircular.

The only analytic issue is whether these tiny cubic cumulant perturbations are free Gibbs laws for strictly convex C^2 potentials. For small a , they are one-cut algebraic perturbations of the variance-1/2 semicircle. Their Hilbert transforms are C^1 -close to the semicircular Hilbert transform, so the associated equilibrium potentials are strictly convex after a harmless extension outside the support.

The counterexample

We use the convention that the Cauchy transform is

$$G_\mu(z) = \int_{\mathbb{R}} \frac{1}{z - x} d\mu(x), \quad z \in \mathbb{C}^+,$$

and that R_μ is determined near zero by

$$z = \frac{1}{G_\mu(z)} + R_\mu(G_\mu(z)).$$

Lemma 1 (Small cubic one-cut perturbations). *For all sufficiently small real a , there is a compactly supported probability measure μ_a whose R -transform is*

$$R_a(w) = \frac{1}{2}w + aw^2.$$

For $a \neq 0$, this measure is not semicircular. Moreover, for $|a|$ small the measure μ_a has one interval of support, a positive real-analytic density in the interior, square-root edge behavior, and boundary Cauchy transform depending C^1 -continuously on a .

Proof. Consider

$$K_a(w) := \frac{1}{w} + \frac{1}{2}w + aw^2.$$

At $a = 0$, the inverse branch of K_0 at infinity is the Cauchy transform of the centered semicircle of variance $1/2$. The two finite real branch points are simple, coming from $K'_0(w) = 0$, and the corresponding density is the usual one-cut square-root density.

For $|a|$ small, the critical equation

$$K'_a(w) = -\frac{1}{w^2} + \frac{1}{2} + 2aw = 0$$

has two simple real critical points perturbing the two semicircular critical points; the remaining critical point is far away and does not affect the inverse branch normalized by $G_a(z) \sim 1/z$ at infinity. The standard algebraic one-cut perturbation argument gives a branch G_a solving

$$z = \frac{1}{G_a(z)} + \frac{1}{2}G_a(z) + aG_a(z)^2$$

which maps $\mathbb{C}^+$ into $\mathbb{C}^-$ and satisfies $zG_a(z) \rightarrow 1$ at infinity. By the Herglotz representation theorem, G_a is the Cauchy transform of a compactly supported probability measure μ_a . Stieltjes inversion gives a positive real-analytic one-cut density with square-root edges, and the construction is analytic in a after rescaling the perturbed support to a fixed interval.

The displayed identity is precisely the defining relation for $R_{\mu_a}(w) = w/2 + aw^2$. If $a \neq 0$, the third free cumulant of μ_a is a , so μ_a is not semicircular. $\square$

Lemma 2 (Convex Gibbs realization). *For all sufficiently small real a , the law μ_a from Lemma 1 is a free Gibbs/equilibrium measure ν_{Φ_a} for some strictly C^2 convex potential $\Phi_a : \mathbb{R} \rightarrow \mathbb{R}$.*

Proof. Let $[\alpha_a, \beta_a]$ be the support of μ_a , and write

$$H_a(x) := \text{p. v.} \int_{\alpha_a}^{\beta_a} \frac{1}{x - y} d\mu_a(y) = \text{Re } G_a(x + i0), \quad x \in (\alpha_a, \beta_a).$$

For $a = 0$, μ_0 is the variance-1/2 semicircle and

$$2H_0(x) = 2x$$

on its support. By the C^1 -continuous one-cut perturbation in Lemma 1, after taking $|a|$ small enough we have

$$\frac{d}{dx}(2H_a(x)) \geq 1 \quad \text{for } x \in (\alpha_a, \beta_a).$$

Choose a C^1 extension $W_a : \mathbb{R} \rightarrow \mathbb{R}$ of $2H_a$ such that

$$W'_a(x) \geq \frac{1}{2} \quad \text{for all } x \in \mathbb{R},$$

and such that outside a compact neighborhood of the support the primitive grows quadratically fast enough. Define

$$\Phi_a(x) := \int_0^x W_a(t) dt + C_a.$$

Then $\Phi_a \in C^2(\mathbb{R})$ is strictly convex.

On the support of μ_a , the Euler-Lagrange equation for logarithmic energy is

$$\Phi'_a(x) = 2 \text{ p. v. } \int \frac{1}{x-y} d\mu_a(y).$$

Equivalently,

$$\Phi_a(x) - 2 \int \log|x-y| d\mu_a(y)$$

is constant on $\text{supp } \mu_a$. By choosing the convex extension outside the support sufficiently steep, the same expression is at least that constant off the support. Hence μ_a is the logarithmic equilibrium measure associated with Φ_a . Strict convexity of Φ_a gives uniqueness, so in the source's notation $\mu_a = \nu_{\Phi_a}$. $\square$

Theorem 1. *The weak free Cramer conjecture from arXiv:2410.02470 is false. There are strictly C^2 convex potentials Φ_+ and Φ_- such that $\nu_{\Phi_+} \boxplus \nu_{\Phi_-}$ is semicircular, while neither ν_{Φ_+} nor ν_{Φ_-} is semicircular.*

Proof. Choose $a \neq 0$ sufficiently small for Lemmas 1 and 2 to apply, and put

$$\mu_+ = \mu_a, \quad \mu_- = \mu_{-a}.$$

By Lemma 2, $\mu_{\pm}$ are free Gibbs measures associated with strictly C^2 convex potentials. Their R -transforms satisfy

$$R_{\mu_+}(z) + R_{\mu_-}(z) = \left(\frac{1}{2}z + az^2\right) + \left(\frac{1}{2}z - az^2\right) = z.$$

Since R -transforms add under free convolution, $\mu_+ \boxplus \mu_-$ has R -transform z , hence is the centered semicircular law of variance 1. But the third free cumulants of μ_+ and μ_- are a and $-a$, respectively, so neither factor is semicircular. $\square$

Verification notes

The algebraic sanity check in `code/verify_cubic_perturbation.py` numerically solves

$$aG^3 + \frac{1}{2}G^2 - zG + 1 = 0$$

for a small value of a , samples the boundary density, and estimates the convexity margin of $2\operatorname{Re}G(x + i0)$ on the support. This is not a substitute for the analytic one-cut perturbation lemma, but it checks the intended branch and the size of the convexity buffer.

Limitations and novelty check

The packet is scoped only to the one-dimensional weak free Cramer conjecture as stated in the source. It does not address the later multidimensional free-transport conjectures or the proposed non-commutative Bakry-Emery curvature criterion.

The bounded novelty check searched the run indexes for arXiv:2410.02470, “Free Stein kernel”, “free Cramer”, “free Gibbs”, and “semicircular”. Web searches for the exact phrase “Weak free Cramer’s conjecture” and for “free Gibbs Cramer’s theorem free convolution semicircular” found the source and the classical Bercovici–Voiculescu failure of free Cramer, but no public answer to the strictly convex free-Gibbs variant in this bounded search.

References

- H. Bercovici and D.-V. Voiculescu, *Superconvergence to the central limit and failure of the Cramer theorem for free random variables*, Probab. Theory Relat. Fields 103 (1995), 215–222.
- C.-P. Diez, *Free Stein kernels and moment maps*, arXiv:2410.02470, 2024.